

Suraj Patel

Lucknow, Uttar Pradesh, India | +91 9260923895 | surajpatel9390@gmail.com

linkedin.com/in/suraj-patel-9201b2381/ | github.com/SurajPatel04

SUMMARY

Dynamic Software Engineer specializing in agentic AI workflows and multimodal RAG platforms, with a proven track record in developing real-time backend systems. Expert in building production-level LLM applications utilizing Python, FastAPI, Node.js, and LangChain, with a strong focus on scalable REST APIs, Redis caching, and cloud deployment. Committed to delivering innovative solutions that enhance performance and user experience.

EXPERIENCE

Careerboat	Dec 2025 - Present
<i>Full Stack Engineer, Remote</i>	
- Reduced LLM token consumption by 40% for the AI interview service by architecting stateful Node.js APIs with MongoDB checkpointing and dynamic context summarization across multi-turn sessions, significantly enhancing efficiency. - Engineered a real-time streaming layer over Server-Sent Events and AWS S3, achieving a time-to-first-audio of approximately 2 seconds, which improved user engagement by delivering token-by-token responses. - Revamped a high-volume job search API by redesigning MySQL FULLTEXT queries and ranking algorithms, slashing search latency from 1-5 minutes to just 5-10 seconds while enhancing search relevance. - Integrated the Razorpay payment gateway for secure transactions, implementing custom backend logic for discount coupons, thereby increasing transaction efficiency.	

EDUCATION

Amity University Online	Jul 2024 - Jun 2026
Master's in Master of Computer Applications (MCA)	
Lucknow Christian College	Jul 2021 - Jun 2024
Bachelor's in Bachelor of Science	

PROJECTS

InsightFlow – Multimodal RAG Platform	insightflow.surajpatel.dev github.com/SurajPatel04/ai-multimedia-rag-app
- Designed a two-phase ingestion flow that ensures uploads are confirmed before processing, effectively eliminating unnecessary embedding costs on unverified files. - Integrated Deepgram for timestamped transcription with clickable media citations, facilitating precise retrieval and playback of media content to the exact second. - Implemented Redis semantic caching for repeated queries and LangGraph-based conversation memory management, achieving a 50% reduction in query load through GCP deployment using Docker and GitHub Actions CI/CD.	
AI Manim Video Generator	video.surajpatel.dev github.com/SurajPatel04/manimVideoGenerate
Engineered a self-healing generation loop with Vision QA, utilizing Vision LLMs (GPT-4o-mini/Gemini) to visually validate rendered frames against user intent, boosting execution success rates from approximately 60% to over 90%.	

- Developed a scalable, asynchronous rendering architecture leveraging Celery, Redis, and Supabase to support concurrent generation, real-time polling, and in-flight task cancellation, secured via JWT and Google OAuth.

SKILLS

- **Languages:** JavaScript, TypeScript, Python
- **Frontend:** React, Redux, HTML5, CSS3
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT, OAuth 2.0
- **Databases & Caching:** PostgreSQL (Prisma), SQL, MongoDB (Mongoose, Beanie), Redis, FAISS (Vector DB)
- **AI:** LangChain, LangGraph, LangSmith (LLM observability), OpenAI API, Gemini API, RAG, Prompt Engineering
- **Tools & DevOps:** AWS (S3), GCP, Docker, Git, GitHub Actions, Linux, Bash/Shell, Celery
- **Core Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design